Replicated: Automated Technical Artifact Analysis

Overview:

Replicated has a tool called Troubleshoot, https://troubleshoot.sh, that creates archive snapshots of Kubernetes cluster data for remote troubleshooting.

An ISV (software vendor) distributes their Kubernetes-based application to an end customer who installs it in a disconnected, self-controlled environment. The end customer operates the application and infrastructure. When something breaks or isn't performing correctly, the ISV developers don't have direct access to the deployed system to debug the issue. The end customer runs the support bundle tool to collect an archive of logs, state, metrics, etc. and sends it to the developers, who analyze the bundle remotely.

The current process requires:

- Manual analysis of potentially large log archives
- Domain expertise to identify relevant signals in logs
- Time-consuming correlation between different log sources
- Knowledge of common failure patterns across K8s, application code, and infrastructure

All without access to a live system to inspect!

At Replicated, we and our vendors handle a lot of these support bundles. The volume and complexity of the task consumes a lot of time. Most of the issues are as simple as a misconfigured value. Others though, will have multiple engineers scratching their heads fruitlessly for hours trying to figure out how the system got in the state that it did.

To that end, we are actively building AI tools to automatically analyze, detect and resolve issues in these support bundles. It is a fascinating endeavor that we expect to be able to *largely automate*.

We think we have a pretty good handle on the whole idea, and we've been having a lot of fun exploring different approaches. We thought it would be interesting for you all as well.

So…

We want you to build a prototype of an AI-enabled system to automatically analyze Troubleshoot support bundles. What it does and how it does it is entirely up to you.

Core Requirements:

You must make a program that takes a Troubleshoot support bundle as input, uses AI to analyze it, and outputs something useful.

Web app, desktop app, tui, bash script, I don't care. Also, this problem is pretty open ended, so there aren't any specific metrics like latency or load times that we will be looking for. We are interested in the breadth and quality of your analysis.

Beyond this, put yourself in our shoes. **What else can you do within the problem domain that would be interesting?**

Technical Constraints:

None. Use any language, framework, application type, LLM or AI tool.

Suggested Approach:

To start, you will, probably, need to:

- Set up a Kubernetes cluster - A local cluster via https://kind.sigs.k8s.io/ or https://minikube.sigs.k8s.io/ works fine. Deploy a simple workload so there's something to collect.
- Install the Troubleshoot support-bundle tool - Installation instructions at https://troubleshoot.sh. It's a kubectl plugin.
- Break something - Introduce some failures in your cluster so your bundle has real issues to detect.
- Generate a support bundle - Run the tool against your cluster to produce a .tar.gz archive containing logs, cluster state, and metrics.

Then…

Expected Deliverables:

- A link to a ~2 minute video demonstrating your analysis system
- A Github repo with code and instructions that I can run against a support bundle. We'll be looking at your repo. Make it presentable.
- A 1 page (500 words max!) markdown file committed in the repo describing your approach to the problem and any interesting thoughts you have about the problem domain, titled MY_APPROACH_AND_THOUGHTS.md